BAKONG

Modernizing Cambodia's Payment Landscape

April 2026

Executive Summary

Cambodia's economy has long faced three structural challenges: heavy dependence on the US dollar, a banking system that reached only a small fraction of the population, and a fragmented domestic payment landscape. In October 2020, the National Bank of Cambodia (NBC), in partnership with the Japanese technology firm SORAMITSU, launched Bakong, a distributed-ledger technology national payment system designed to address all three.

Bakong is built on Hyperledger Iroha, a permissioned distributed ledger framework. It connects over fifty participating banks, microfinance institutions, and payment service providers within a unified national infrastructure. The system is accessible through traditional bank accounts (backbone model) as well as a dedicated Bakong App developed by NBC (extended access). This dual-access design lowers entry barriers and allows unbanked users to join the platform directly through their mobile phones.

Adoption has been further accelerated by KHQR, the domestic QR code standard. KHQR encodes the recipient's account information and removes the need to enter account details manually when making a payment, streamlining the process and improving the user's experience.

The system, which processed the equivalent of 4.5 times the Cambodian GDP in 2025 , has been expanding internationally and formally joined the ASEAN Regional Payment Connectivity (RPC) initiative in April 2025. It already enables cross-border[1] QR payments with eight countries (Thailand, Laos, Vietnam, China, Malaysia, South Korea, Japan, and Singapore) with further links to India, Rwanda, Fiji, and the Solomon Islands in development, alongside connectivity through Alipay+, WeChat, UnionPay, Visa and Mastercard.

This paper documents the vision, architecture, and implementation of Bakong, as well as the practical challenges encountered along the way. It is offered as a reference and case study for policymakers, financial institutions, and central banks engaged in digital payment modernization in emerging economies.

INTRODUCTION

The way money flows through an economy reflects how that economy is organized. For most of Cambodia's modern history, value moved through cash, predominantly USD paper notes, passed hand to hand, recorded in fragmented ledgers, and settled across banks operating in silos. The economy functioned but lacked the efficiency required to support its rapid development: moving money was slow and costly, rural communities were underserved, and monetary sovereignty steadily eroded to a foreign currency.

This was not for lack of effort. Since the turn of the millennium, the National Bank of Cambodia (NBC) has driven a sustained financial inclusion agenda, restructuring the banking sector under the Law on Banking and Financial Institutions, and later embracing mobile payments, lifting the financial inclusion rate from 3.7% in 2011 to 39% in 2024. Based on recent NBC primary data, this rate could even reach up to 73% in 2026. Yet payment infrastructure remained fragmented, and dollarization ran deep, with roughly 90% of banking sector credits and deposits denominated in foreign currency.

To address this situation, the NBC launched the Bakong project, designed as a structural response and developed in partnership with SORAMITSU, a Japanese technology provider. Bakong processes transactions in both riel and US dollars and is accessible to any Cambodian or foreign visitor with a mobile phone, acknowledging Cambodia's dollarization while creating the conditions for riel promotion and broader financial inclusion. The approach is working, between 2021 and 2025, local currency transactions grew to over half of all transactions, accounting for roughly one-third of total transaction value.

The architecture chosen for Bakong reflects this dual objective. NBC adopted a distributed model that allows participating institutions to settle directly with one another in near real time. A transfer between a customer of a small microfinance institution and a customer of a large commercial bank clears within seconds, at low cost, and with finality. This brings institutions of different sizes and backgrounds onto common infrastructure and extends to rural users the same settlement capability previously available mainly to urban clients.

Significant challenges remain. Dollarization is still deep, and the share of transaction value denominated in dollars shows that the riel's gains have so far been concentrated in smaller, everyday payments. Continued progress will depend on sustained macroeconomic stability and deeper use of formal financial services. The direction is clear: value that once moved as cash passed hand to hand, recorded in fragmented ledgers and settled across banks operating in silos, now flows through shared infrastructure in which the riel and the dollar stand on equal technical footing, and in which the organization of the Cambodian economy, and its money, are no longer fragmented by design.

THE CAMBODIAN CONTEXT: THREE STRUCTURAL CHALLENGES

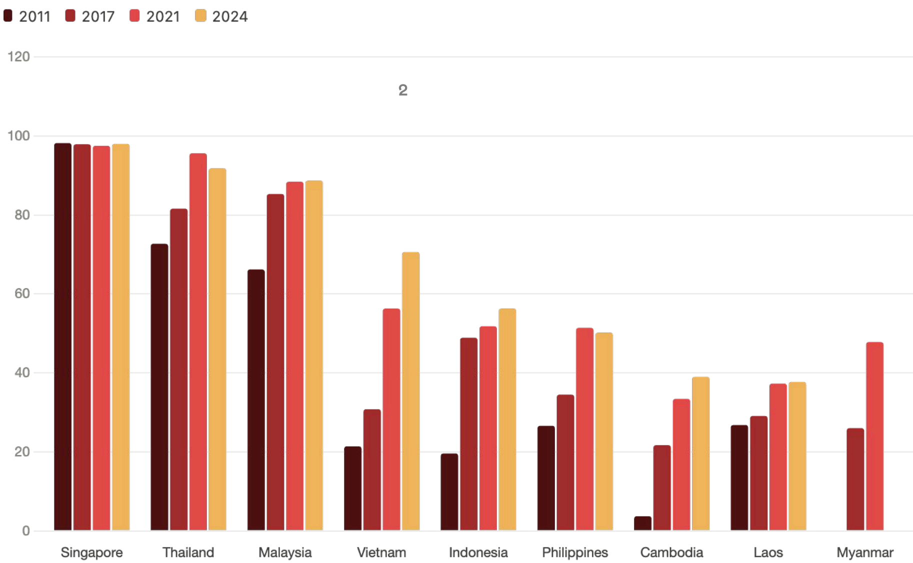
Bakong was designed in response to three structural challenges facing the Cambodian economy: a highly dollarized monetary system, a largely unbanked population, and a fragmented payment landscape.

A Highly Dollarized Economy

Beginning in the late 1980s and accelerating through the 1990s, the US dollar circulated widely in Cambodia as a hedge against political and monetary instability. Decades later, despite a stable exchange rate, low inflation, and sustained growth, the conditions generally considered conducive to de-dollarization, the dollar still accounts for approximately 90% of bank deposits, loans, and transaction value. Dollarization provided stability during a period of recovery, but it also constrains the NBC's ability to conduct independent monetary policy, manage liquidity, and respond to domestic economic shocks.

Bakong was designed to operate in both riel and US dollars, bridging the present reality of a dollarized economy with the longer-term objective of riel promotion. By giving the national currency a digital channel as convenient as the dollar, the system supports a gradual transition.

An Unbanked Population



Proportion of adults aged 15+ with financial account in ASEAN economies, 2011 - 2024 (%) (World Bank's Global Findex database 2011, 2017, 2021 and 2024)

Much of the Cambodian economy remains rural and informal, offering limited commercial incentive for banks to serve these segments, and the physical distance between branches and rural communities further discouraged formal-sector participation. Account ownership was correspondingly low: according to Global Findex, only 3.7% of adults held a formal account in 2011, rising to 39% by 2024 but still below the regional average. NBC primary data suggests this rate could reach as high as 73% in 2026. In 2020, there were 3,520,742 loan accounts (equivalent to 4,204,382 borrowers), a figure that rose to 4,521,887 loan accounts (5,131,198 borrowers) by 2025. In that context, wages, retail transactions, and remittances largely flowed through cash and informal channels.

Reaching these populations fell to the NBC under its financial inclusion mandate. The angle adopted was mobile-first: phone penetration was already near-universal, providing a channel into households the formal banking system had not. Bakong was built on this premise, accessible directly through a mobile application without requiring a preexisting bank account, a credit history, or a branch visit.

A Fragmented Payment Landscape

By the late 2010s, Cambodia hosted a growing number of banks, microfinance institutions, and payment service providers, yet customers of one institution could not easily transact with those of another. The country had no real-time gross settlement system: interbank clearing ran on a twice-daily batch cycle, so a payment instructed after the cut-off waited until the next business day. Worse, clearing-house access was restricted to commercial banks holding settlement accounts at the NBC, excluding the payment service providers that served the underserved. The effect was distributional: a parent in the provinces could not conveniently remit to a child studying in the capital, and fragmentation hardened an existing divide between urban and rural, banked and unbanked.

A second fragmentation followed commercial logic. Each institution ran its own wallet and app in pursuit of a dominant position, with little incentive to interoperate, much as, in Cambodia's early telecom market, subscribers on one network could not call another. Interoperability survived only at the merchant level, where shopkeepers displayed several QR codes side by side to accommodate their customers. Distributed ledger technology offered the way out: its immediate, irrevocable settlement finality eliminates the risk that made regulators wary of admitting payment service providers, allowing every licensed institution to interoperate within one settlement framework. Bakong embedded that interoperability as a feature of the national infrastructure rather than a burden carried by merchants, and the KHQR standard replaced a proliferation of proprietary codes with a single format accepted across the system.

The quick adoption of payment QR codes in Cambodia

BAKONG ARCHITECTURE AND DESIGN

This section sets out the technical foundations of Bakong: the distributed ledger framework, the network topology, and the privacy model. Several intrinsic characteristics of the underlying technology made it the right foundation for the system:

- Simple onboarding. Users open an account with their regular bank or download the Bakong App to register directly. Technically, onboarding generates a public/private key pair for each user.
- Security. Bakong is operated by the National Bank of Cambodia and distributed through locally regulated banks, offering a high level of safety for Cambodian users' assets.
- Gross atomic settlement. In a distributed-ledger system, the ledger update and the transaction message are a single gross operation: funds and messaging move together, with no separate clearing step. This eliminates the time gap between payment, clearing and settlement found in conventional systems, and, combined with a gross settlement model, removes interbank settlement risk by construction.

The Window for Change

The convergence of widespread mobile penetration and a largely underserved population created the opportunity for a structural upgrade of Cambodia's national payment infrastructure.

NBC began researching distributed ledger payment systems in 2016 and formalised its development partnership with SORAMITSU, a Japanese technology provider, in 2017. A live pilot ran from July 2019 with more than 10,000 users, paving the way for full production launch on 28 October 2020. KHQR followed in July 2022 as the unified domestic QR standard, marking the shift from a fragmented payment system to a coherent national payment ecosystem.

The adoption of KHQR also opened the path to cross-border interoperability. Because KHQR follows the EMV specification which is a global technical standard for secure, chip-based credit and debit card payments, it can be mapped to other compliant national QR standards through bilateral arrangements. Cross-border integration began shortly afterwards with Thailand and extended in subsequent years to Vietnam, Laos, Malaysia, China, South Korea, and Japan. Cambodia formally joined the ASEAN Regional Payment Connectivity initiative in April 2025, anchoring Bakong within the region's emerging payments architecture.

Project Bakong | Key Milestones

2016
DLT Research Begins

2017
SORAMITSU Partnership

2019
Live Pilot (>10k Users)

2020
Production Launch (Oct)

2022
KHQR National QR Standard

2023
Cross-border Expansion

2025
ASEAN RPC Initiative (Apr)

Bakong project timeline, from initial DLT research in 2016 to ASEAN Regional Payment Connectivity in 2025.

Overview: the Bakong layered system

Bakong can be summarized as a three-tier architecture, with each layer assigned a distinct role. At the foundation sits the distributed ledger that records and secures every transaction; above it, the Bakong Core orchestrates the system and enforces its rules; and at the top, participating institutions connect through their own payment gateways to reach end users. The three levels are described below, from top to bottom.

Level 3 - Top Participant Institutions & Payment Gateway

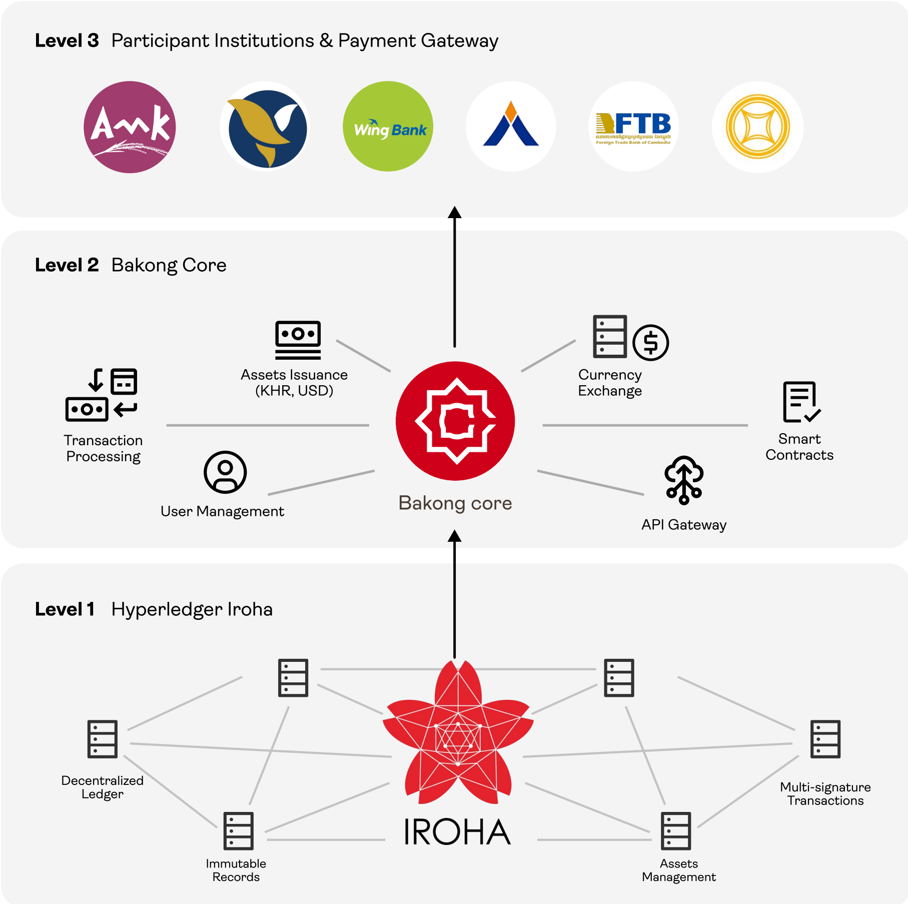
Payment gateways operated by participating payment service providers: commercial banks, microfinance institutions (MFIs), and e-wallet providers. Each institution operates its own gateway, which integrates with the Bakong Core APIs and makes the payment service available to end users through its own application: ACLEDA Mobile, FTB Bank, Wing Bank, Vatanac Bank and others.

Level 2 - Middle Bakong Core

The Bakong Core, operated by the National Bank of Cambodia, is the system's orchestration layer: it manages member accounts, validates transactions against business rules, exposes APIs to participants, and writes settled transactions to the distributed ledger (which is operated exclusively by NBC nodes).

Level 1 - Bottom Hyperledger Iroha

Hyperledger Iroha is a permissioned distributed ledger that records every transaction. It is the trust layer of Bakong: it guarantees the immutability of the data and ensures consensus among the NBC's validator nodes.



Architecture of Bakong with 3 levels

Hyperledger Iroha, characteristics and topology

The core ledger of Bakong is built on Hyperledger Iroha, an open-source permissioned distributed ledger framework hosted by the Linux Foundation.

The choice of innovation

Iroha was selected for several properties relevant to a central-bank-operated payment system: it is permissioned, with every participant approved by the NBC; it offers Byzantine fault-tolerant consensus, eliminating double-spend risk without proof-of-work mining; it provides fine-grained role-based access control; and it is designed for high-throughput retail use.

The Bakong core runs on physical infrastructure operated by the NBC. Identical copies of the ledger are replicated across institutional nodes, all operated by the NBC on ring-fenced hardware, providing redundancy against both hardware failure and tampering. Consensus among these distinct nodes ensures that no single one can unilaterally alter transaction history, reinforcing the integrity of the infrastructure. Reported throughput exceeds 2,000 transactions per second, with end-to-end settlement typically completed in under five seconds.

Privacy by Design

A defining feature of Bakong's design is the separation of transaction data from personally identifiable information (PII). The core ledger stores transactions, but it does not store names, identification numbers, or other personal data. Personally identifiable information is held by the financial institutions that onboarded each user. Each institution sees only the transactions in which one of its customers is a counterparty, not the system's full traffic.

This separation gives the NBC system-wide visibility through aggregated data collection for monetary policy (nowcasting potential), anti-money-laundering (pattern analysis), and macroprudential purposes (liquidity flow), while preserving meaningful user privacy and keeping institutions' customer data compartmentalized.

The two Bakong models

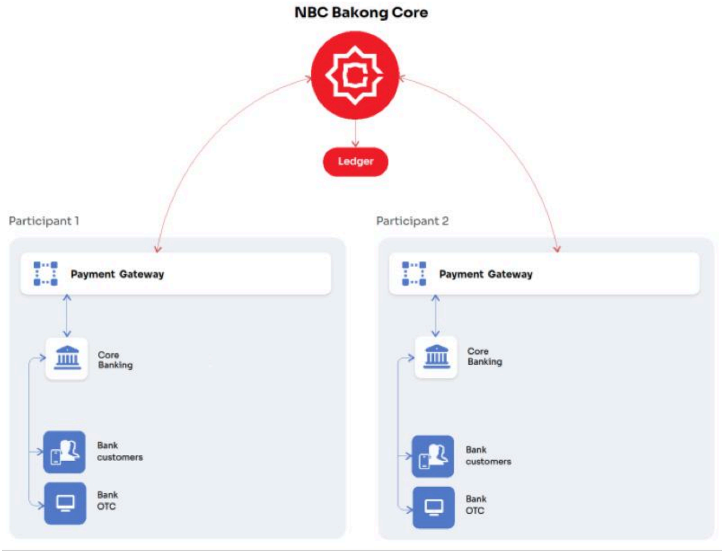
On top of this technology architecture, Bakong offers two access models. The first is the classic intermediated model, called backbone: Bakong serves as the backbone of the national payment system, connecting commercial banks and payment service providers, with end users accessing it through their regular accounts with their own financial institution. The second is the extended model: end users can also connect directly to the payment gateway of a participating institution through the Bakong App, without holding a prior account.

The Backbone model

Through the backbone model, Bakong connects commercial banks and payment service providers within a shared national payment infrastructure. Each participant continues to serve its end users through its own application and usual distribution channels, with customer funds recorded in current accounts held on those institutions' books. Payments and transfers between institutions are routed and settled through Bakong, enabling seamless credit transfers across the network. As of today, around 8.5 million users can access the Backbone model through their existing bank accounts.

The system operates on a real-time gross settlement basis: each transaction is prefunded, settled individually, and final on execution. Because Bakong is built on a distributed ledger, messaging and the movement of funds occur in a single operation. Payments are instant, settlement risk is eliminated, and reconciliation across participants is streamlined.

The backbone sits alongside Cambodia's other national payment infrastructure: the National Clearing System (NCS) for batch and cheque clearing and settlement, and the Cambodia Shared Switch (CSS) for domestic card transactions. Together, these three systems form the layered architecture of Cambodia's national payment system.



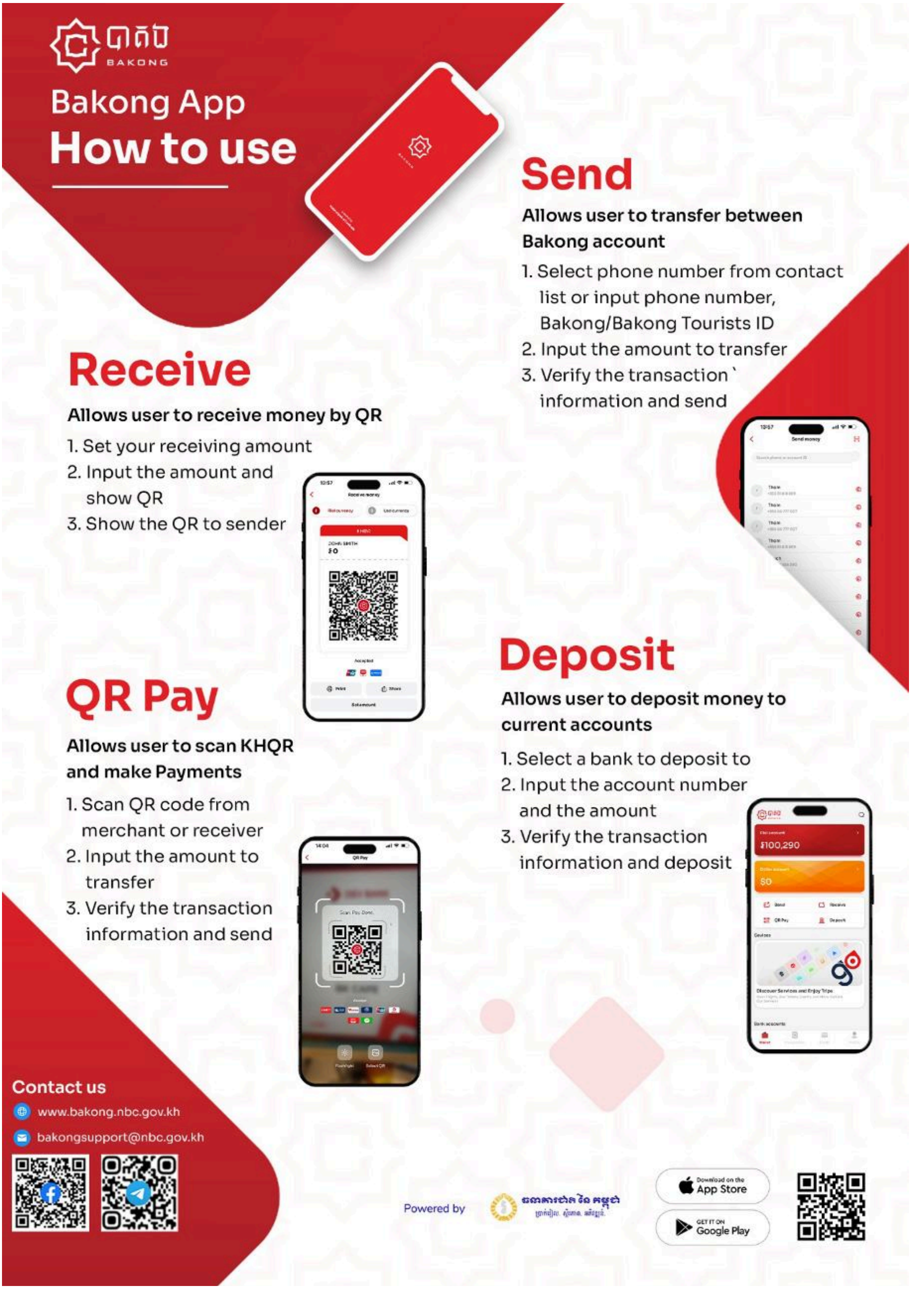
Bakong Backbone architecture

Bakong App, or the Extended model

Complementing the Backbone model, NBC developed the Bakong App as a direct-access channel into the system, called the Extended model. Currently, there are about 700,000 Bakong App active users. This second model serves two purposes:

Extending Bakong's reach to the unbanked, particularly in rural areas. Without holding a prior bank account, a user can download the app and complete KYC with an eligible participating institution hosting the customer subwallet (NBC itself among them) and immediately send, receive, and hold funds in both Khmer riel and US dollars,

Leveling the playing field across the banking sector. When Bakong was first introduced, not all participating institutions offered their own mobile applications to customers. The Bakong App provided these institutions with a ready-made mobile interface, as well as desktop access for corporate users, ensuring that no participant was excluded from the system for lack of a digital front end.

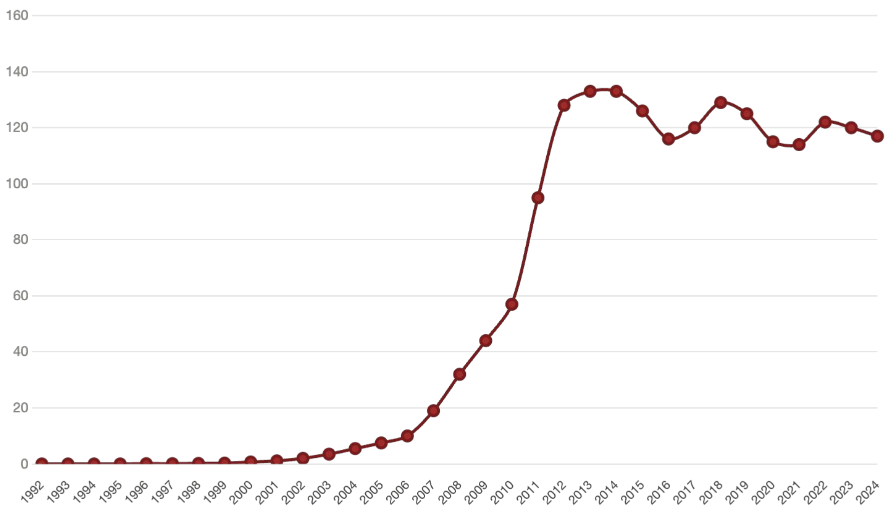


Main functions of Bakong: Send, Receive, QR Pay and Deposit

Reaching the users, Bakong's end goal

A central design choice defines Bakong: to reach end users directly and indirectly, rather than only providing the backbone alone.

NBC's earlier FAST system had followed the conventional path, meaning supplying interbank rails and leaving each institution to build its own customer-facing application. At the time Bakong launched, only the largest banks had done so, leaving the customers of smaller institutions with no digital channel and no access to digital payment solutions. By offering the Bakong App as a ready-made front end, NBC closed that gap, ensuring that no participant and no customer was excluded for want of a digital interface. As mobile development costs have since fallen and most participants have built their own apps, the App's role has evolved: from a gap-filler at launch to an inclusive on-ramp for the still-unbanked.



Mobile adoption in Cambodia constantly over 100% since early 2010's (World Bank Data, Serie Mobile cellular subscriptions per 100 people – Cambodia,).

KHQR

Layered across the two Bakong models is KHQR, the unified QR standard that ensures every participant can transact with every other. Together, these three components form the core of a digital public infrastructure supporting Cambodia's economic development. Three deliberate design choices explain the rapid adoption of the KH QR in Cambodia, now equipping 4.5 million acceptance points.

The first is interoperability. Before KHQR, each bank had begun issuing its own QR code, which only shifted the fragmentation problem to merchants: shopkeepers ended up displaying several codes side by side at the point of sale to accommodate their customers' preferred apps. KHQR is the single EMV-compliant QR standard for payments in Cambodia: every participant's application can read and process every other participant's QR code. A merchant displays one code, and a customer pays with whichever participating app they prefer.

The second is user experience. KHQR encodes the recipient's account information directly, so the payer no longer needs to enter a mobile number, account number, or account name to complete a transaction. Scanning replaces typing, reducing both friction and error. Payment becomes instantaneous, a significant driver of customer adoption.

The third is commercial sustainability. Because KHQR runs directly on Bakong, funds reach recipients instantly. Crucially, the standard is free of charge for both users and merchants, and merchants receive payments in full, unlike other digital payment channels, which deduct fees. These mechanics create a strong incentive for merchants to adopt KHQR, while the standard's interoperability lets banks compete on equal terms to onboard them. The result is a virtuous circle: merchant demand pulls in banks, and bank competition drives further KHQR and Bakong uptake.

ID	Length	Value	Meaning
00	02	01	Payload Format Indicator
01	02	11	Point of Initiation Method (dynamic QR)
29	13	0009dave@pras	Merchant Account Information
52	04	5999	Merchant Category Code
53	03	116	Currency Code (116 = KHR)
58	02	KH	Country Code
59	06	SOSORO	Merchant Name
60	10	Phnom Penh	Merchant City
63	04	3FDA	CRC Checksum

KHQR code: a complete payment instruction in a single scan.

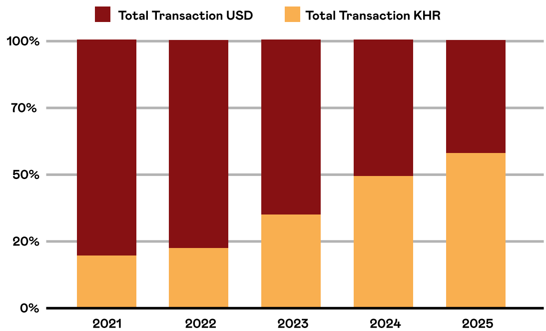
THE BAKONG SOLUTION

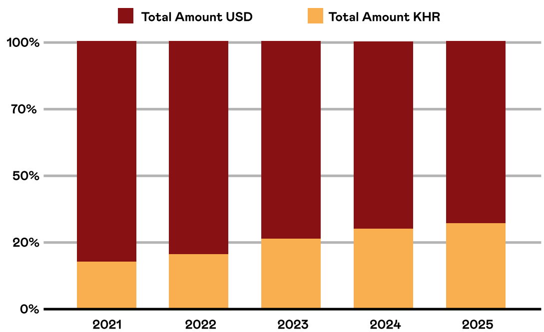
Bakong and De-Dollarization

For the first time, a single system let users hold and transact the riel digitally across the full range of financial institutions and across both urban and rural populations. The effect on local currency use was measurable: between 2021 and 2025, riel transactions grew to over half of all transactions by volume and one-third by value. This was the result of two deliberate design choices:

The first is in-app currency flexibility. Bakong wallets are dual-currency, and users convert between riel and dollars in real time at their bank's exchange rate, turning foreign exchange into a function of the payment interface rather than a trip to a branch or exchange counter.

The second, and more consequential, governs cross-border payments: outbound cross-border transactions can only be initiated from a riel account, and merchants accepting payments from tourists must likewise hold one. For users and merchants who previously held only dollars, this creates a direct commercial incentive to open a riel account and for merchants, the reward is access to the entire cross-border customer base. The NBC has thus embedded de-dollarization into the architecture of the payment system itself, working through market incentives rather than regulatory mandate.





Growth of local currency usage in Bakong transactions (2021–2025)

Bakong and Financial Inclusion

Bakong advanced financial inclusion on two fronts. It reaches rural and unbanked users directly, with simple registration and basic functions that open access beyond the reach of commercial banks. And it accelerated the sector itself: in 2019 only a handful of banks offered a mobile app; today every bank in Cambodia does. In 2025, Bakong reaches 8.5 million users with 4.5 million KHQR acceptance points.

The deeper impact is data. Every payment, a wallet transfer, a remittance, a QR transaction, leaves a verifiable financial record. For individuals and small businesses long excluded from formal finance, these records accumulate into a profile of revenue, cash flow, and reliability that institutions can use to extend credit and other products. Bakong widens inclusion not through subsidy, but by generating financial visibility.

Round-the-clock availability completes the shift. Unconstrained by branch hours, financial services move from a privilege of physical proximity to an on-demand utility available to anyone with a mobile phone.

From big-box retail to the corner stall, Cambodia pays by phone.

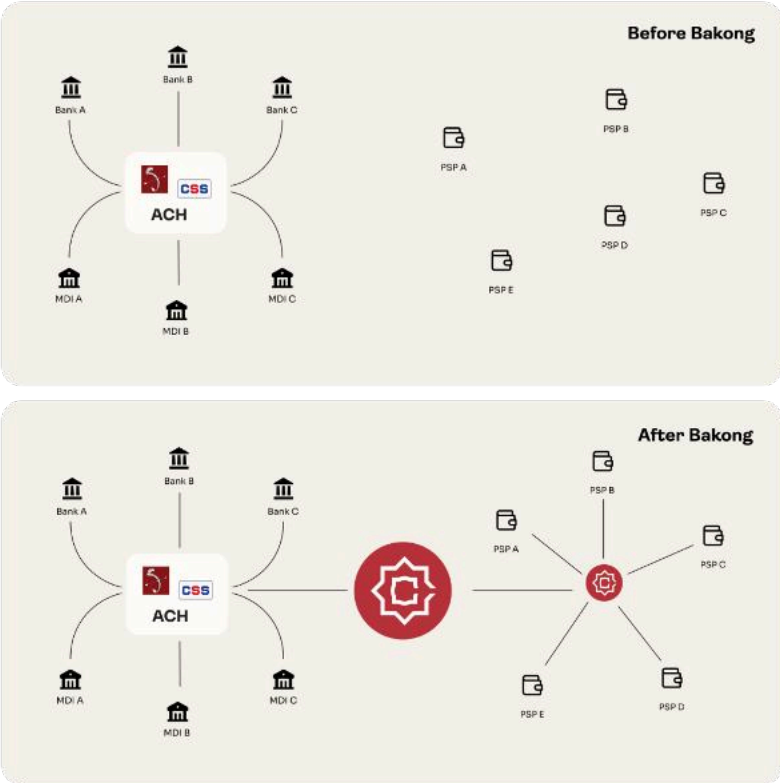
Bakong and Interoperability

The fragmentation described earlier — institutions unable to transact with one another, merchants displaying competing QR codes — was resolved by Bakong at two distinct layers, and it is worth setting them out together. Bakong made every player in Cambodia's financial system interoperable, and it does so across both.

From a user perspective, the first is the KHQR layer, which sits on top and does two things at once. For users, it standardises the experience: one QR format is read and processed by every participating app, so any customer can pay any merchant. For institutions, it standardises the data: a KHQR code resolves to an address within Bakong, which then routes the funds to the recipient's account. This shared, structured format is what makes participation easy to implement, as institutions code to one specification rather than negotiating formats with every counterpart.

At the backend, the second is the settlement layer, the Bakong Core and the distributed ledger beneath it. This is the settlement platform itself: every bank and payment service provider connects to a single core system, and transactions between any two participants settle there instantly, with finality and no settlement risk. Interoperability is no longer something institutions must negotiate bilaterally, it is a property of the infrastructure they all share. This achieves what balance-sheet interconnections between institutions could not deliver on their own: where domestic reach previously depended on bilateral correspondent links and selective club arrangements, available to some participants and not others, and on terms set privately between them, Bakong extends the same settlement capability to every participant equally, regardless of size or standing.

Together, the two layers separate concerns cleanly: KHQR makes the system easy to use and easy to build on, while the Bakong settlement layer guarantees that payments can be processed, safely and swiftly.



Payment Landscape in Cambodia post and pre- Bakong payment system

The naga of Angkor Wat, a multi-headed serpent, symbolizes protection and the link between humans and gods. It represents power, water, and the ancient union of the naga princess and the Indian prince, the mythical origin of the Khmer people.

INTERNATIONAL DEVELOPMENT

Bakong Tourists

Building on Bakong's success, the National Bank of Cambodia has extended the system to international visitors through Bakong Tourists, bringing the same seamless, cashless payment experience enjoyed by locals to travellers. For a country where tourism is a major source of foreign earnings, this is a key challenge to address.

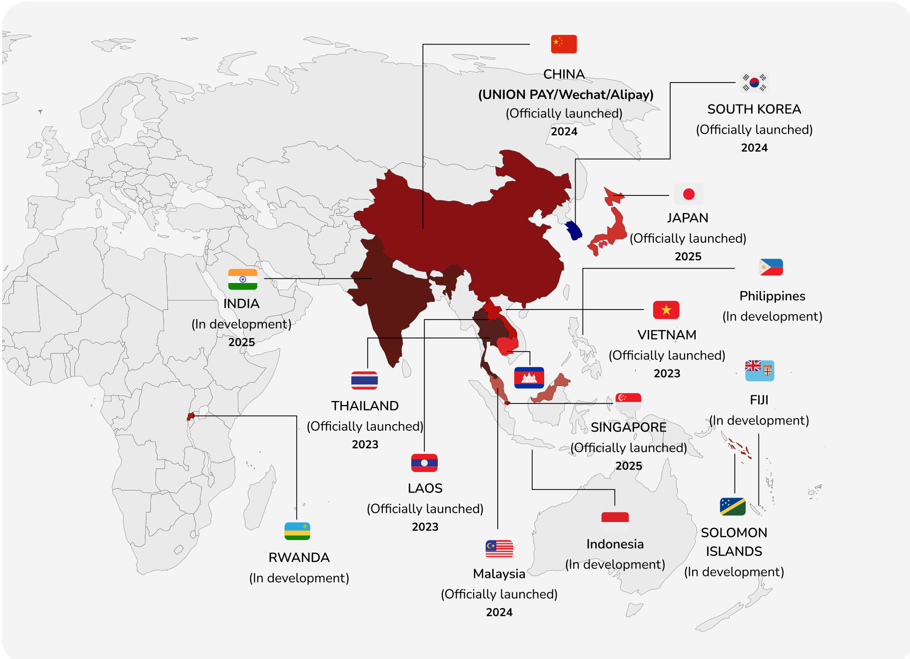
On Bakong Tourists, registration requires only an email address, and the wallet can be topped up with a Visa or Mastercard, or with cash at partner banks. Balances are held in both US dollars and Khmer riel, with real-time conversion between the two, so visitors can pay in the currency a merchant expects without managing the exchange themselves. By scanning a KHQR code, visitors can pay at over 4.5 million acceptance points nationwide, from street-food stalls and tuk-tuk drivers to markets, cafés, and hotels. Easier to implement and operate, QR codes have a far wider reach than the roughly 42,000 card POS terminals across the country. Beyond convenience, this removes the need to carry cash, a real gain in safety.

Bakong Tourists: a payment solution for visitors to Cambodia

Cross-Border Reach

Bakong's international footprint now spans eight operational links – Thailand, Laos, Vietnam, China (via UnionPay International, AliPay+, and WeChat Pay), Malaysia, South Korea, Japan, and Singapore – with four further integrations in development: India, Rwanda, Fiji, and the Solomon Islands.

Bakong's development and international expansion have received broad recognition. Awards from Central Banking Publications, together with case studies published by the World Economic Forum, the Linux Foundation Decentralized Trust, and the Asian Development Bank, have positioned Bakong as a leading and innovative reference design for payment system modernisation in developing economies.



Bakong's cross-border network: operational linkages, and integrations in development.

CHALLENGES AND FORWARD AGENDA

Bakong did not launch perfect. Designed to improve people's lives, it has had to adapt to problems encountered along the way. Several challenges shaped, and continue to shape, its development.

Ecosystem management

An emerging payment system is only as strong as the network around it. Bakong's technology was necessary but not sufficient: its success depended equally on the legal framework it operates within and on the active participation of the financial institutions that connect to it. Three dimensions proved decisive.

Oversight

Existing laws, regulations and controls had to be reviewed to confirm that the new system was adequately covered, with gaps identified and updated where necessary. A central-bank-operated distributed-ledger system was not initially anticipated in NBC's frameworks: settlement finality, the legal standing of the system, and the inclusion of payment service providers previously excluded from clearing-house access all had to be reviewed and clarified, along with the extension of oversight and anti-money-laundering rules to the new participants. This groundwork was a precondition for launch and an ongoing work to accompany its development, domestically and abroad.

Go-to-Market

Bakong faced a go-to-market challenge on two fronts: it had to win the participation of financial institutions and the trust of the public. The first depended on positioning Bakong as a complement, not a competitor. The Bakong App is deliberately limited to basic payment and transfer functions, leaving the customer relationship and any value-added services to the participants, who remain free to innovate and differentiate in their own apps. Reassured that the national infrastructure would not disintermediate them, banks and payment service providers had a clear incentive to join. On the demand side, adoption was driven by the user experience itself — instant, free, interoperable payments through a single QR standard — which gave the public a reason to use the system and, in turn, gave merchants a reason to accept it.

Integration and maintenance

As a backbone, Bakong depends on each participant integrating it into their own systems and then keeping that integration running: maintaining the connection, applying updates, and allocating the human and material resources needed for smooth day-to-day operation. This is an ongoing commitment, not a one-off build, and it cannot simply be mandated. Sustained engagement with participants was therefore essential, ensuring that institutions saw enough value in the system to keep investing in it well after launch.

Technical issues and user experience

Building and running a national payment system raises challenges that extend well beyond launch day. Three in particular have shaped Bakong's ongoing development: reaching users with limited connectivity or digital literacy, safeguarding infrastructure that is now critical to the economy, and detecting and resolving fraud as transaction volumes grow.

Fraud detection and resolution

As Bakong scales and connects to payment systems abroad, it also widens the surface on which fraud and scams can occur, from social-engineering schemes targeting individual users to coordinated cross-border activity. Instant, irrevocable settlement, one of the system's core strengths, also means there is little room to reverse a fraudulent transfer once executed, which places a premium on prevention, rapid detection, and clear resolution procedures.
This is not work the NBC can carry alone. Effective fraud management depends on participating banks and payment service providers, which hold the customer relationship and the front-line monitoring tools; on coordination with foreign authorities and operators as cross-border links multiply; and on cooperation with law enforcement and other domestic agencies.

Rural connectivity and digital literacy

Adoption remains uneven across the country. Areas with weaker mobile coverage and lower digital literacy still show meaningfully lower uptake, and without deliberate effort the system risks tracking, rather than closing, an existing divide. Narrowing it calls for sustained outreach and infrastructure upgrades on one side, and product design on the other: offline functionality for intermittent coverage, voice-based interfaces for users who do not read, and a lightweight version of the app for less capable handsets. The goal is to ensure that the populations Bakong was created to reach are not the last to benefit from it.

Operational and cyber risk

A national payment backbone is, by its nature, critical infrastructure, and the consequences of failure are systemic. The distributed-ledger architecture mitigates several classes of attack — there is no single point of control to compromise — but no system is fully immune. The NBC continues to invest in monitoring, incident response, and consensus-node diversity to protect the integrity of the core system. Because the backbone is only as resilient as the institutions connected to it, the NBC also strengthens banking supervision, ensuring that participants hold their own critical IT systems to a comparable standard.

CONCLUSION

Bakong stands as a concrete example of a central bank deploying distributed ledger technology to address structural national challenges on a realistic timeline. NBC built a workable system, deployed it, and learned along the way, rather than waiting for a fully optimized design.

Five years on, the results are tangible:

- A meaningful share of Cambodia's transaction value now moves through a single, interoperable, central-bank-operated infrastructure.
- The riel has, for the first time at scale, a digital channel more convenient than paper dollars, supporting the gradual rebalancing of the country's monetary base.
- Financial inclusion has advanced through two reinforcing channels: direct access for the unbanked through the Bakong stand-alone App, with around 700,000 active accounts to date and reaching 8.5 million users through the Bakong backbone, as well as 4.5 million KHQR acceptance points and a broader behavioural shift in which the convenience of mobile payments has driven users to open bank accounts to participate in the new payment norm.

Bakong did not only bank the unbanked, it created a new national payment culture that deepened the formal financial system. The work is not finished: adoption must extend further into rural areas, riel use must keep growing alongside the dollar, and cross-border integration is still expanding.

Its longer-term significance lies in what a trusted, interoperable national payment infrastructure makes possible once in place.

Several of these prospects may look like side products, yet each carries core benefits. A rail that reaches citizens directly gives the government a channel for delivering public services (social transfers, subsidies, pensions) and a more transparent means of collecting taxes. The same traceability that supports anti-money-laundering analysis can, applied to public funds, improve the reliability of state spending. And an infrastructure that settles value with finality is a foundation for deeper financial markets: as tokenized securities gain ground internationally, Bakong gives Cambodia a settlement layer already in operation rather than still to be designed. In each case the pattern holds: a payment system, once trusted and universal, becomes general-purpose public infrastructure whose return compounds well beyond payments.

As a reference architecture for emerging-market payment modernization, Bakong now stands as an operational and documented case study. Its lessons are likely to inform central bank thinking well beyond Cambodia's borders.

ធនាគារជាតិនៃកម្ពុជា

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